

Zhengwei Liu

Data science / Time series analysis / Bayesian statistics

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Research Interests

Applications Traffic data modeling and analysis; analysis of underreported infectious disease data.

Methods and theory Count time series, Gaussian processes, dynamic linear models, semiparametric models, panel quantile regression, and network time series.

Professional Experience

Nanjing Normal University, Lecturer

Jul 2021 - Present

School of Mathematical Sciences

Education

Jilin University

Sep 2016 - Jun 2021

Ph.D. in Probability and Mathematical Statistics

Jilin University

Sep 2012 - Jun 2016

B.Sc. in Mathematics and Applied Mathematics

Research Grants (Principal investigator)

Jiangsu Provincial Natural Science Foundation, Young Scientists Program

Sep 2024 - Present

Modeling and inference for count time series with underreporting

Jiangsu Provincial Department of Education, General Program

Jul 2022 - Jun 2024

Statistical inference and prediction for Bayesian median autoregressive models

Publications (* denotes the corresponding author)

- Guo, X., **Liu, Z.**, & Zhu, F. * (2026). Time-varying dispersion negative binomial network autoregression. *Acta Mathematica Sinica, English Series*, accepted.
- Liu, Z.**, & Zhu, F. * (2026). A hybrid quantile regression for fixed effects panel data. *TEST*, 35(2), 559-589.
- Liu, Z.**, & Zhu, F. * (2024). Asymmetric exponential power Bayesian median autoregression with applications. *Journal of Statistical Computation and Simulation*, 94(9), 2063-2086.
- Liu, Z.**, Li, Q., & Zhu, F. * (2021). Semiparametric integer-valued autoregressive models on $\mathbb{Z}$. *Canadian Journal of Statistics*, 49(4), 1317-1337.
- Liu, Z.**, & Zhu, F. * (2021). A new extension of thinning-based integer-valued autoregressive models for count data. *Entropy*, 23(1), 62.
- Liu, Z.**, Li, Q., & Zhu, F. * (2020). Random environment binomial thinning integer-valued autoregressive process with Poisson or geometric marginal. *Brazilian Journal of Probability and Statistics*, 34(2), 251-272.

Preprint

- Fan, Y., **Liu, Z.** *, & Zhu, F. (2026). An efficient method of posterior sampling for Poisson INGARCH models. *arXiv*, 2603.07527.

Talent Program

2022 | Jiangsu Province Shuangchuang Doctoral Talent Program